

ASTRID HERREMANS

Celestijnenlaan 200A, 3001 Leuven, Belgium

`astrid.herremans@kuleuven.be`

RESEARCH INTERESTS

Numerical analysis at the intersection of approximation theory and numerical linear algebra (non-orthogonal expansions, sampling theory, rational approximation), using tools from harmonic analysis, functional analysis and random matrix theory.

EDUCATION

PhD in Engineering Science 2022–2026

KU Leuven, Department of Computer Science

- Advisor: Daan Huybrechs
- Expected date of submission: June 2026
- Teaching in linear algebra, analysis, partial differential equations, numerical analysis and fundamentals of computer science
- Ombudsperson for the Master in Mathematical Engineering

Master in Mathematical Engineering 2020–2022

KU Leuven

- Grade: summa cum laude with the congratulations of the examination committee
- Includes advanced courses in numerical analysis, numerical linear algebra, numerical simulation of PDEs, nonlinear systems and scientific software (Fortran, C++)
- Thesis: *Robust and Efficient Approximation of Functions with Singular Behaviour*, advised by Daan Huybrechs
- Teaching in linear algebra and analysis

Bachelor in Engineering Science 2017–2020

KU Leuven

- Grade: summa cum laude
- Provides a broad foundation in Engineering Sciences with a chosen focus on Computer Science, including courses in programming (Python, Java) and core mathematical subjects

PUBLICATIONS AND PREPRINTS

Reproducible code is made available on my GitHub, primarily in Julia and MATLAB.

A. HERREMANS AND B. ADCOCK. *Refinement-based Christoffel sampling for least squares approximation in non-orthogonal bases*. arXiv:2510.08461 (2025).

A. HERREMANS AND D. HUYBRECHS. *Function approximation in numerically rank-deficient bases*. arXiv:2501.07470 (2025).

A. HERREMANS AND D. HUYBRECHS. *Efficient function approximation in enriched approximation spaces*. IMA J. Numer. Anal. 45.2 (2025), 673-695.

N. BOULLÉ, A. HERREMANS AND D. HUYBRECHS. *Multivariate rational approximation of functions with curves of singularities*. SIAM J. Sci. Comput. 46 (2024), A3401-A3426.

A. HERREMANS, D. HUYBRECHS AND L.N. TREFETHEN. *Resolution of singularities by rational functions*. SIAM J. Numer. Anal. 61.1 (2023), 2580-2600.

TALKS AND SEMINARS

A selection of my slides can be found on my homepage (astridherremans.eu).

Foundations of Computational Mathematics (FoCM) at the University of Vienna, invited by Teresa Perez, Erik Koelink and Daan Huybrechs	July 2026
NUMA seminar at KU Leuven	March 2026
Applied Mathematics Seminars at the University of Pavia, invited by Umberto Zerbinati and Andrea Moiola	November 2025
The 30th Biennial Numerical Analysis Conference at the University of Strathclyde, invited by Marcus Webb and Timon Gutleb	June 2025
Numerical Analysis Group Internal Seminar at the University of Oxford, invited by Yuji Nakatsukasa	June 2025
Challenges, Opportunities, and New Horizons in Rational Approximation plenary talk at the Banff International Research Station (BIRS) invited by Heather Wilber, Anil Damle and Serkan Gugercin	April 2025
PIMS MOCAD seminar at Simon Fraser University, invited by Ben Adcock	March 2025
Monte Carlo and Quasi-Monte Carlo Methods in Scientific Computing at the University of Waterloo (contributed talk)	August 2024
More on Harmonic Analysis in Strobl (poster session)	June 2024
SIAM Conference on Applied Linear Algebra at Sorbonne University, invited by Nadiia Derevianko and Heather Wilber	May 2024
Workshop and Summer School on Applied Analysis at TU Chemnitz (contributed talk)	September 2023
Numerical Analysis in the 21st Century in honour of Nick Trefethen's retirement from Oxford at the University of Oxford (contributed talk)	August 2023
Classical Analysis Seminar at the department of Mathematics at KU Leuven, invited by Arno Kuijlaars	March 2023
SIAM Conference on Computational Science and Engineering in Amsterdam (contributed talk)	February 2023
Numerical Analysis Group Internal Seminar at the University of Oxford, invited by Lloyd Nick Trefethen	November 2022

AWARDS AND GRANTS

SIAM UKIE prize for best Student Talk by SIAM UKIE <i>Awarded at the 30th Biennial Numerical Analysis Conference</i>	2025
Travel grant for a long stay abroad by Research Foundation Flanders (FWO) <i>Covers a three-month research stay at Simon Fraser University, hosted by Ben Adcock</i>	2025
SIAM Student Travel Grant by SIAM <i>Covers travel and accommodation costs for the SIAM Conference on Applied Linear Algebra</i>	2024
PhD fellowship by Research Foundation Flanders (FWO) <i>Awarded a fully funded four-year PhD fellowship for fundamental research</i>	2023

Best contributed talk

2023

by the Faculty of Mathematics at TU Chemnitz

Awarded at the Workshop and Summer School on Applied Analysis

Best Student Award

2018

by KU Leuven

Awarded to the top ten students in the first year of the Bachelor of Engineering Science